

Simplify Algebraic Expressions

Lesson 6-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.


$$3x + 2x = 5x$$

same variable → combine

5x and 3x are like terms (both x); 5x and 5y are NOT (different variables); 2x and 2x² are NOT (different powers)

Like Terms

Write the definition:

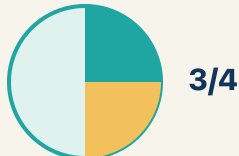

$$2 + 3 = 5$$

$$2 + 3 = 5$$

8m + 3m = 11m — add the coefficients (8 + 3 = 11) and keep the variable (m)

Combine

Write the definition:



2a + 5a + 3 simplifies to 7a + 3 — the two a-terms combine, the constant stays

Simplify

Write the definition:


$$4x$$

coefficient = 4

In 9x, the coefficient is 9 — it tells you 'nine groups of x'; if x = 2, then 9x = 18

Coefficient

Write the definition:

$$3x + 7 + y$$

3 separate terms

In $3x + 5 - 2x$, the terms are $3x$, 5 , and $2x$.

Term

Write the definition:

$$3(x + 2)$$

$$3x + 6$$

$$3(x + 4) = 3 \cdot x + 3 \cdot 4 = 3x + 12.$$

Distributive property

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can simplify algebraic expressions by combining like terms.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property

1

Terms with the same variable part, like $3y$ and $8y$, are _____.

2

To add or subtract like terms into a single term is to _____ them.

3

To write an expression in its shortest equal form is to _____ it.

4

The number multiplied by a variable is the _____.

5

A number or variable separated by $+$ or $-$ signs is a _____.

6

Multiplying each part of a sum by a number uses the _____ property.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Simplify $7x + 3x$.

A. $10x$

B. $10x^2$

C. $21x$

D. $73x$

Step 1 $7x + 3x = 10x$.

Step 2 Add the coefficients: $7 + 3 = 10$.

 **Answer:** A. $10x$

 We do — together

Solve this one with your class using the same steps.

Problem: Simplify $9n + 4 + 2n$.

A. $11n + 4$

B. $15n$

C. $11n^2 + 4$

D. $92n + 4$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which pair are like terms?

A. $6x$ and $2x$

B. $6x$ and $6y$

C. $6x$ and 6

D. $6x$ and $2x^2$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Clean up this channel: $7y - 3y$.

- A. $4y$
- B. $10y$
- C. 4
- D. $21y$

Show your work:

2. Simplify the mix: $6m + 4 + 3m$.

- A. $9m + 4$
- B. $13m$
- C. $9m + 4m$
- D. $10m$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Sentence starter: Student ____ is correct because $4x + 2x + x = \underline{\hspace{1cm}}x$ and $3 + 5 = \underline{\hspace{1cm}}$. Student ____'s error was _____. Student ____'s error was _____.

Show your work:

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

- A. $8x + 8$
- B. $8x^2 + 8$
- C. $62x + 35$
- D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **We Try (notes):** A. $11n + 4 - 9n + 2n = 11n$. *The constant 4 has no like term, so it stays.*
Result: $11n + 4$.
8. **You Try (notes):** A. $6x$ and $2x - 6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).
9. **Try It 1:** A. $4y$ — Subtract the coefficients of the like terms: $7 - 3 = 4$. Keep the variable y , giving $4y$.
10. **Try It 2:** A. $9m + 4 - 6m + 3m = 9m$. *The constant 4 has no like term, so it stays. Result: $9m + 4$.*
11. **Exit Ticket:** A. $8x + 8 - 6x + 2x = 8x$ and $3 + 5 = 8$, so the simplified expression is $8x + 8$.